ECON 144 — EXTENSIVE R CODE REFERENCE ADF TEST (Unit Root) library(tseries) adf.test(x) # H0: unit root KPSS TEST (Stationarity) library(urca) kpss.test(x) # H0: stationary LJUNG-BOX TEST (Autocorrelation) Box.test(x, lag=12, type="Ljung-Box") # H0: no autocorrelation ARCH TEST library(FinTS) ArchTest(x) # H0: no ARCH effects GARCH MODEL library(rugarch) spec <- ugarchspec(variance.model = list(model="sGARCH", garchOrder=c(1,1)), mean.model = list(armaOrder=c(0,0))) fit <- ugarchfit(spec, data=x) COINTEGRATION: ENGLE–GRANGER reg <- lm(y ~ x) res <- reg$residuals adf.test(res) # H0: no cointegration ECM ESTIMATION u <- residuals(lm(y ~ x)) ecm <- lm(diff(y) ~ diff(x) + lag(u, -1)) KALMAN FILTER library(dlm) mod <- dlmModPoly(order=1, dV=R_value, dW=Q_value) filtered <- dlmFilter(y, mod) MARKOV CHAIN STATIONARY DISTRIBUTION library(expm) P <- matrix(c(0.7,0.3,0.2,0.8),2,2,byrow=TRUE) pi <- steadyStates(P) GIBBS SAMPLING (TWO NORMAL CONDITIONALS) theta1 <- theta2 <- numeric(N) for (t in 2:N) { theta1[t] <- rnorm(1, mean=theta2[t-1], sd=1) theta2[t] <- rnorm(1, mean=theta1[t], sd=1) } DIEBOLD-MARIANO TEST library(forecast) dm.test(e1, e2, alternative="two.sided") # H0: equal accuracy